

IMMUNE AND LYMPHATIC SYSTEMS



EVERY DAY THE BODY IS INVADDED by disease-causing micro-organisms called pathogens, or germs. The immune and lymphatic systems are the body's defence against these pathogens. The immune system is a collection of cells that keep detailed records of invading pathogens so, if they reappear, they can be destroyed, making you immune to that disease. The lymphatic system drains fluid called lymph from tissues, filters out any pathogens, and returns the lymph to the bloodstream.

Lymph nodes

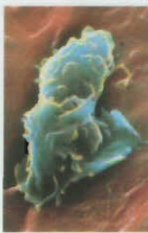
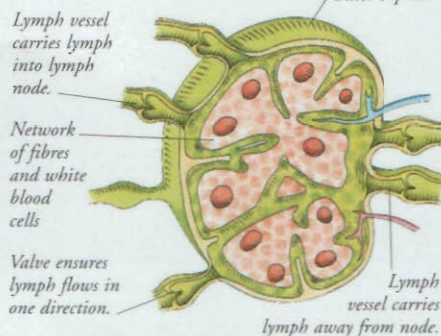
Lymph constantly leaves the bloodstream and flows through the spaces surrounding cells. It passes through lymph nodes, which are small swellings of the lymph vessels that clean and filter the lymph. Inside each lymph node is a network of fibres which supports large numbers of two types of immune system cells, lymphocytes and macrophages.



Lymphocytes

Lymphocytes, found in the lymph nodes recognize and destroy specific pathogens using chemicals that are called antibodies. Lymphocytes are also to be found circulating within the bloodstream.

Cutaway of a lymph node



Macrophages

Macrophages are cells with voracious appetites that detect, engulf, and destroy viruses, bacteria, cancer cells, and any other foreign material in the lymph that passes through the lymph node.

Tonsils guard throat against infections.

Neck lymph nodes

Thoracic duct empties lymph into a main vein.



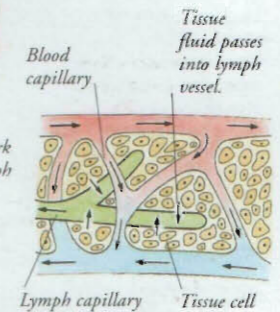
Lymphatic system

The lymphatic system consists of a network of tubes called lymph vessels that reach all parts of the body, and several lymphatic organs. Lymph is carried by the vessels to the main lymph ducts which empty into the bloodstream.

Heart

Spleen is a lymph organ that also stores blood.

Lymph node removes pathogens from lymph.



Lymph vessels

The smallest lymph vessels are capillaries. Excess fluid drains through the walls of lymph capillaries from the surrounding tissue. Lymph capillaries join to form larger lymph trunks.



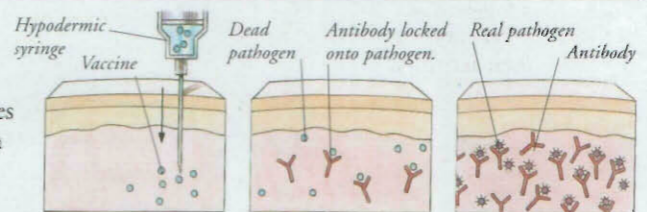
Quilt for AIDS charity

AIDS

Acquired Immune Deficiency Syndrome (AIDS) is caused by the Human Immunodeficiency Virus (HIV). A person with AIDS becomes infected with diseases that the body would normally fight off. This is because HIV attacks and destroys immune system cells. In time, the immune system weakens and the person becomes unable to fight infections and eventually dies.

Immunization

Immunization gives a person protection against a specific disease. There are two types of immunization. In active immunization, a vaccine containing some dead pathogens is injected into the body to stimulate the immune system to make antibodies. Passive immunization involves injecting antibodies and gives short-term protection.



1 A vaccine containing dead or weakened pathogens is injected into the body.

2 The immune system produces antibodies and keeps a "memory" of the pathogen.

3 If the real pathogens enter the body, large numbers of antibodies are released.

Lady Mary Wortley Montagu

Lady Mary Wortley Montagu (1689–1762) was an English author who introduced an early form of immunization against smallpox to England. In Turkey, she had seen people scratching pus from small-pox blisters into the skin of healthy people to protect them from catching smallpox. She had her children "vaccinated" and publicized the method.



Allergies

If you have an allergy, it means that your immune system has wrongly identified a harmless substance, called an allergen, as being harmful. The body's reaction to these allergens produces symptoms such as sneezes and rashes. Common allergens include pollen, fur, dust, shellfish, and strawberries.

Strawberry



A skin patch test is used to identify possible allergens.

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